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INTERNSHIP TECHNICAL REPORT

NISAR SATELLITE DATA PRODUCTS AND BHOONIDHI PORTAL

Technical Study & Practical Demonstration

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MACROEDTECH

August 2026

TABLE OF CONTENTS

ABSTRACT

The NASA-ISRO Synthetic Aperture Radar (NISAR) mission is an important Earth observation mission developed by NASA and ISRO. It uses L-band and S-band radar to collect information about the Earth's surface. Unlike optical satellites, radar can collect data even when there are clouds or during the night. NISAR data can be useful for studying agriculture, forests, floods, natural disasters, land surface changes, water bodies, urban areas and other environmental conditions. This report gives a basic study of the NISAR mission, its data products, applications and data processing methods. The Bhoonidhi Portal was also studied to understand how NISAR data can be searched, viewed, accessed and downloaded. Different steps of working with NISAR data were also studied, from data collection to map preparation. For the practical part of the work, NISAR S-band HHHH data was used for a selected urban area of Kolkata. The downloaded data was processed and converted into GeoTIFF format for further analysis. The radar backscatter values were studied and converted into decibel (dB) values. Based on the backscatter values, a simple classification was carried out to separate urban and non-urban areas. The classified result was then cleaned to reduce small unwanted pixels and improve the map.

Finally, the urban classification was combined with the NISAR backscatter map to prepare an overlay map. This helped to understand the relation between radar backscatter and urban areas. The practical work provided basic experience in handling NISAR SAR data and using GIS software for remote sensing analysis. The study shows that NISAR data can be useful for different Earth observation and GIS applications. In the future, the work can be improved by using data from different dates, additional radar polarizations, machine learning methods and other satellite data. Multi-temporal NISAR data can also be used to study urban growth and land surface changes over time.

INTRODUCTION:

Satellite data are now widely used for studying and monitoring different changes on the Earth's surface. Remote sensing helps us to collect information about the Earth without directly visiting the area. It is useful in many fields such as agriculture, forest monitoring, disaster management, environmental studies, land-use analysis and other geospatial applications. Synthetic Aperture Radar (SAR) is an important type of remote sensing technology that uses microwave signals to observe the Earth's surface. One of the main advantages of SAR is that it can collect information during both day and night and is less affected by cloud cover compared to optical satellite data. Because of this, SAR data are useful for monitoring surface changes and different natural and environmental processes. The NASA-ISRO Synthetic Aperture Radar (NISAR) mission is a joint project of NASA and the Indian Space Research Organisation (ISRO). The mission uses two different radar frequency bands, L-band and S-band, to observe the Earth's surface. NISAR data can be useful for studying land deformation, agriculture, forests, natural hazards, ecosystems and several other Earth-related processes.

For this study, the NISAR mission and its different data products are examined to understand their characteristics and possible uses. The Bhoonidhi Portal is also explored to understand how satellite data can be searched, viewed, accessed and downloaded. Along with the technical study, a practical demonstration using NISAR data is planned to understand how the data can be used in a real geospatial application. The main purpose of this report is therefore to understand the NISAR mission, its data products, applications and processing workflow, and to explore the role of the Bhoonidhi Portal in satellite data access. The study also considers possible future applications of NISAR data in geospatial analysis, environmental monitoring and other research areas.

OBJECTIVES OF THIS STUDY:

The main objectives of this study are to understand the NISAR mission, its data products and their uses in Earth observation and geospatial studies. The study also focuses on exploring the Bhoonidhi Portal to learn how satellite data can be searched, visualized and accessed. In addition, a practical demonstration using NISAR data will be carried out to understand its use in a real geospatial application. The study also aims to document the analysis, results and source code for future reference and research.

- To learn about the NISAR mission and its main purpose in Earth observation.
- To understand the different NISAR data products and their basic uses.
- To study the major applications of NISAR data in different fields.
- To understand the basic steps of NISAR data processing and analysis.
- To explore the Bhoonidhi Portal and understand its main features.
- To study the available NISAR S-band SAR data through the Bhoonidhi Portal.
- To carry out a simple practical analysis using NISAR data.
- To understand the main results obtained from the practical analysis.
- To document the code and practical work in a GitHub repository.
- To identify some possible future uses of NISAR data.

2: OVERVIEW OF THE NISAR MISSION

2.1 Introduction to NISAR

NISAR stands for NASA-ISRO Synthetic Aperture Radar. It is an Earth observation mission developed by NASA and ISRO. The main purpose of this mission is to observe the Earth's surface and study changes that happen over time. NISAR uses radar technology to collect information about the Earth's surface. It can observe different types of land and environmental features, such as forests, agricultural areas, water bodies and land surfaces. The data can also help in studying changes caused by natural events such as floods, landslides and earthquakes. One useful feature of NISAR is that radar observations can be collected during both day and night. They are also less affected by cloud cover than optical satellite observations. Because of this, NISAR data can be useful for regular monitoring of the Earth's surface. The mission carries two radar systems, L-band and S-band, which provide information at different radar frequencies. The data from these systems can be used for scientific research as well as different practical applications.

2.2 NASA–ISRO Collaboration

NISAR is a joint mission of NASA and the Indian Space Research Organisation (ISRO). Both organisations have contributed important technologies and systems to the mission. NASA is responsible for the L-band SAR system, while ISRO provides the S-band SAR system. The two organisations have also worked together on other important parts of the mission, including the satellite platform and related ground systems. This collaboration combines the experience and technology of both organisations. It also supports the collection of useful Earth observation data for scientific studies and practical applications.

2.3 Mission Objectives

The main objective of NISAR is to observe and understand changes on the Earth's surface. The mission is designed to collect radar data that can be used for studying different natural and environmental processes. Some important areas of study include land deformation, forests, agriculture, water resources, glaciers and natural hazards. NISAR observations can also help in monitoring changes caused by floods, landslides and earthquakes. Another important purpose is to collect repeated observations of the same areas. By comparing observations from different times, researchers can identify changes on the Earth's surface and study them more effectively.

2.4 NISAR Satellite and SAR Technology

NISAR uses Synthetic Aperture Radar (SAR) technology for Earth observation. SAR sends microwave signals towards the Earth's surface and records the signals that are reflected back to the satellite. The information from these signals can be processed to understand different properties and changes of the surface. Unlike optical sensors, SAR does not depend on sunlight, so it can collect data during both day and night. SAR is also useful in areas with frequent cloud cover because microwave signals can pass through clouds more effectively than visible and near-infrared light. This makes SAR suitable for continuous monitoring of many areas. NISAR uses two radar frequencies, L-band and S-band, which provide complementary information about the Earth's surface.

2.5 L-band and S-band SAR

NISAR carries two different SAR systems: L-band SAR and S-band SAR. The L-band SAR is provided by NASA, while the S-band SAR is provided by ISRO. These two bands have different wavelengths and interact with surface features in different ways. The L-band can be useful for studying vegetation, land surface and other features where longer-wavelength radar signals are helpful. The S-band provides additional information about the Earth's surface and vegetation. Using both radar bands together allows NISAR to collect a wider range of information. This is one of the important features of the mission.

<u>3: NISAR DATA PRODUCTS</u>

NISAR collects different types of data from the Earth’s surface. These data are processed into different data products. Each product provides different information and can be used for different purposes.

3.1 Introduction to NISAR Data Products

NISAR data products are the different types of data obtained from the NISAR satellite after collecting and processing radar signals. The data goes through different processing steps before it becomes useful for users. Different products are made for different purposes. Some products contain basic radar information, while other products provide more processed information that can be directly used for analysis and research.

3.2 Level-0 Products

Level-0 products contain the basic data received from the radar system of the satellite. This is the starting form of the data before detailed processing is done. It contains the information collected by the radar during the observation. Level-0 data is mainly useful for further processing and for users who need to work with the original radar information.

3.3 Level-1 Products

Level-1 products are created by processing the basic radar data. The data is converted into a more useful form so that it can be used for SAR image analysis. These products provide information about the Earth's surface and can be used to identify different surface features and changes. Level-1 data is useful for many remote sensing and geospatial studies.

3.4 Level-2 Products

Level-2 products contain more processed information compared to the earlier levels. These products can provide information related to different properties of the Earth's surface. They can be useful for studying soil, vegetation, land surface changes and other environmental conditions. Such products are helpful for scientific studies and different practical applications.

3.5 Other Relevant Data Products

NISAR also provides some additional data and information that help users understand and use the main data products. These supporting data can be useful during data processing, analysis and interpretation. They help users work with NISAR data more easily and correctly.

3.6 Product Characteristics and Data Formats

NISAR data products have different characteristics depending on the type of data and its processing level. Important characteristics include spatial resolution, coverage, data size and file format. Spatial resolution tells us how much detail the data can show on the Earth's surface. Coverage tells us about the area observed by the satellite. Different data products may also have different file sizes and formats. These characteristics are important because they help users choose the right NISAR data product for their study and analysis.

3.7 Applications of Different Data Products

NISAR data products can be used for many purposes. They can help in forest monitoring, agriculture, flood mapping, landslide studies and disaster management. They are also useful for studying soil, vegetation and changes in the land surface. NISAR data can support environmental studies, scientific research, GIS and remote sensing work.

4. NISAR Data Processing Workflow

NISAR collects radar data from the Earth's surface using L-band and S-band systems. The collected data goes through several steps before it can be used for analysis. First, the data is accessed and downloaded, then it is pre-processed and corrected. After that, the SAR data is processed to extract useful information. Finally, the results are prepared as maps or other products for different applications. The main steps of the NISAR data processing workflow are shown below. . These products can be useful for agriculture, forest monitoring, disaster management and environmental studies. The main steps of the NISAR data processing workflow are shown below.

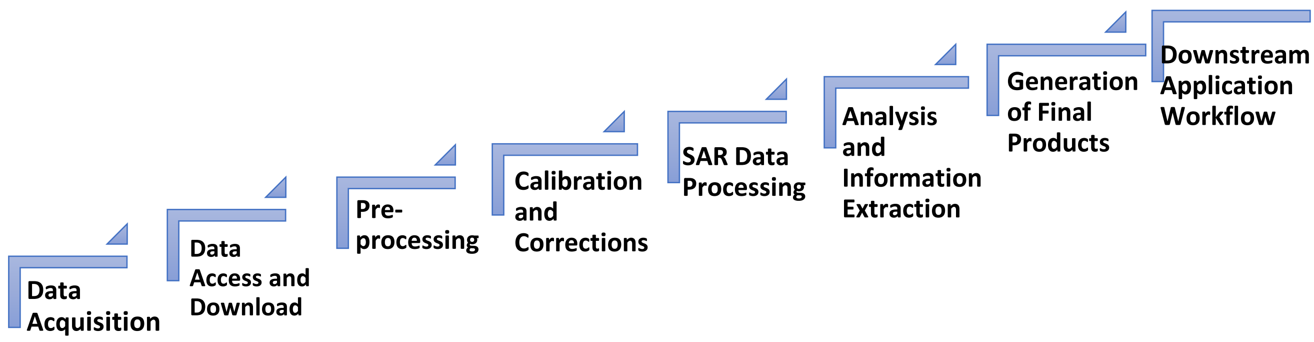

Figure4.1: General workflow of NISAR data processing from data acquisition to downstream applications.

- **Data Acquisition:** NISAR collects data from the Earth's surface using radar. It uses L-band and S-band systems to collect this data.
- **Data Access and Download:** The required NISAR data is searched through the data portal. The selected data can then be downloaded for further work.
- **pre-processing:** The downloaded data is prepared before analysis. Some basic steps are done to make the data ready for use.
- **Calibration and Corrections:** Some corrections are made to improve the quality of the data. This also helps to reduce errors.
- **SAR Data Processing**: The SAR data is processed to make it useful for analysis. Different methods can be used depending on the study.
- **Analysis and Information Extraction:** The processed data is studied to get useful information. It can show changes in land, vegetation and other surface features.
- **4.7 Generation of Final Products:** The final results can be prepared as maps, images or other outputs. These can be used for further study.
- **4.8 Downstream Application Workflow:** The final data can be used for agriculture, forest monitoring, flood studies, disaster management and environmental studies.

5: Applications of NISAR Data

NISAR data can be used in many areas to study changes on the Earth's surface. Some important applications are given below.

- **5.1 Agriculture and Forest Monitoring**: NISAR data can help monitor crops, vegetation and forest areas. It can also help in studying changes in agricultural land and forests.
- **5.2 Flood and Disaster Management**: NISAR data can be used to monitor floods and other natural disasters. It can help identify affected areas and support disaster management.
- **5.3 Land and Surface Change Monitoring**: NISAR can help detect changes in the land surface. It can be useful for studying land deformation and landslides.
- **5.4 Urban and Environmental Monitoring**: NISAR data can be used to study changes in urban areas and the environment. It can also support monitoring of coastal areas and other natural features.
- **5.5 Scientific Research**: NISAR data can support research in remote sensing, GIS, geology, agriculture, environmental studies and other Earth science fields.

6: Exploration of the Bhoonidhi Portal

6.1 Introduction to Bhoonidhi Portal

Bhoonidhi is an Earth Observation data portal developed by the National Remote Sensing Centre (NRSC), ISRO. The portal provides access to satellite data and related Earth observation resources. It allows users to search for satellite datasets based on location, satellite, sensor, resolution, product type and other parameters. For this study, the Bhoonidhi Portal was explored with special attention to the NISAR S-band SAR data products. The portal was studied to understand how satellite data can be searched, visualized, accessed and downloaded for further geospatial analysis.

6.2 Bhoonidhi Homepage

The Bhoonidhi homepage provides access to different satellite data and related services. The major sections of the portal include data browsing and ordering, NISAR information, satellite-related resources, visualization facilities and other data access services. The homepage acts as the main entry point for users who want to explore and access Earth observation datasets.

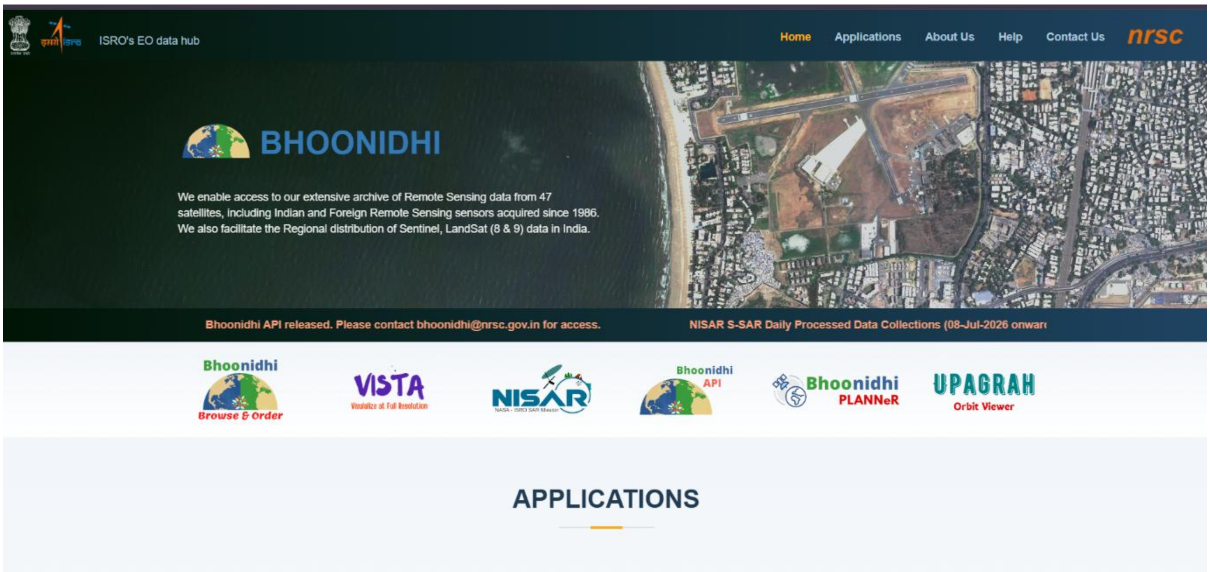

Figure 6.2: Bhoonidhi Portal Homepage

Source: Bhoonidhi Portal, NRSC/ISRO

6.3 Browse and Order

The Browse and Order section is one of the main components of the Bhoonidhi Portal. It allows users to search for satellite data according to their area of interest and required data characteristics. The search interface provides several options such as:

- Place Name
- Latitude and Longitude
- Area of Interest
- Satellite

- Sensor Type
- Resolution
- Product
- Theme
- Open or Priced data

These filtering options help users identify suitable datasets without searching the complete satellite archive manually.

Figure 6.3: Browse and Order Interface of Bhoonidhi Portal

Source: Bhoonidhi Portal, NRSC/ISRO

6.4 Searching Satellite Data

Users can search for satellite data by selecting a place or area. They can also use latitude and longitude to find data. Different filters can be used, such as satellite, sensor, resolution and product. For NISAR data, users can select the NISAR satellite and S-band SAR data. This helps users find the data needed for their study.

6.5 NISAR Section of Bhoonidhi

The NISAR section of the Bhoonidhi Portal provides information about the NISAR mission. It also gives information about NISAR data products and related resources. This section was studied to understand the available NISAR S-band SAR data.

Figure 6.5: NISAR Section of Bhoonidhi Portal

Source: NISAR – NASA-ISRO SAR Mission

6.6 NISAR Product Search

The NISAR product search option helps users find NISAR data for a particular area and date. During this study, NISAR S-band SAR data was searched through the Bhoonidhi Portal. The available products were checked before selecting the data for analysis.

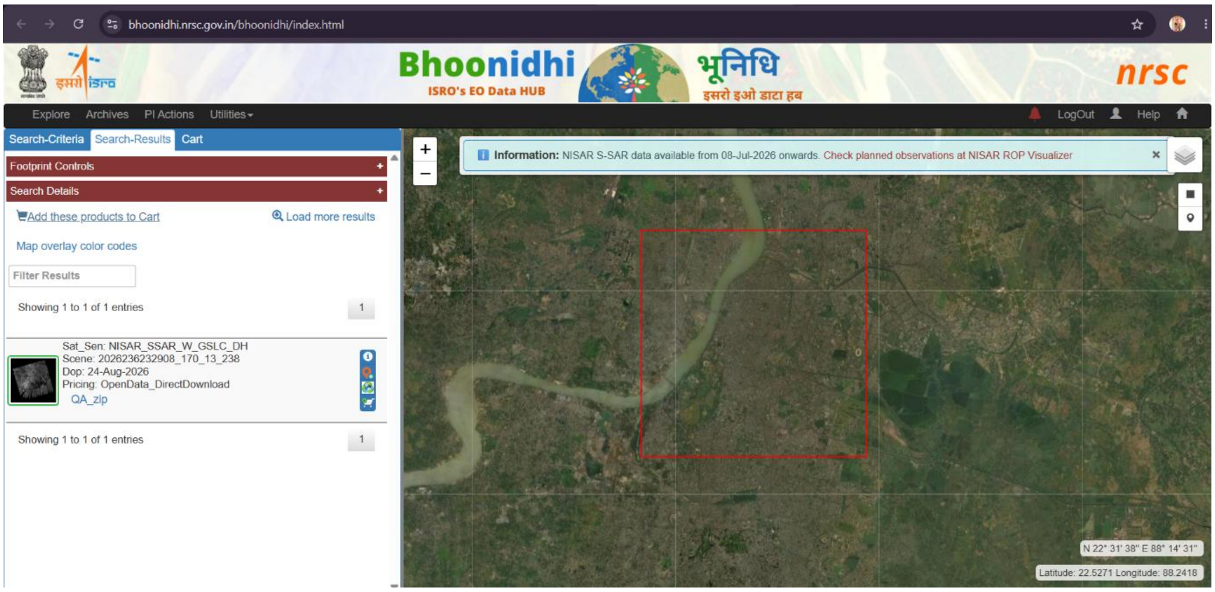

Figure 6.6: NISAR S-SAR Product Search Result in Bhoonidhi Portal

Source: Bhoonidhi Portal, NRSC/ISRO

6.7 Product Visualization

The portal allows users to view the selected satellite data before downloading it. This helps users check the area covered by the data. This is useful because NISAR data files can be very large. Checking the data first helps users select the correct product.

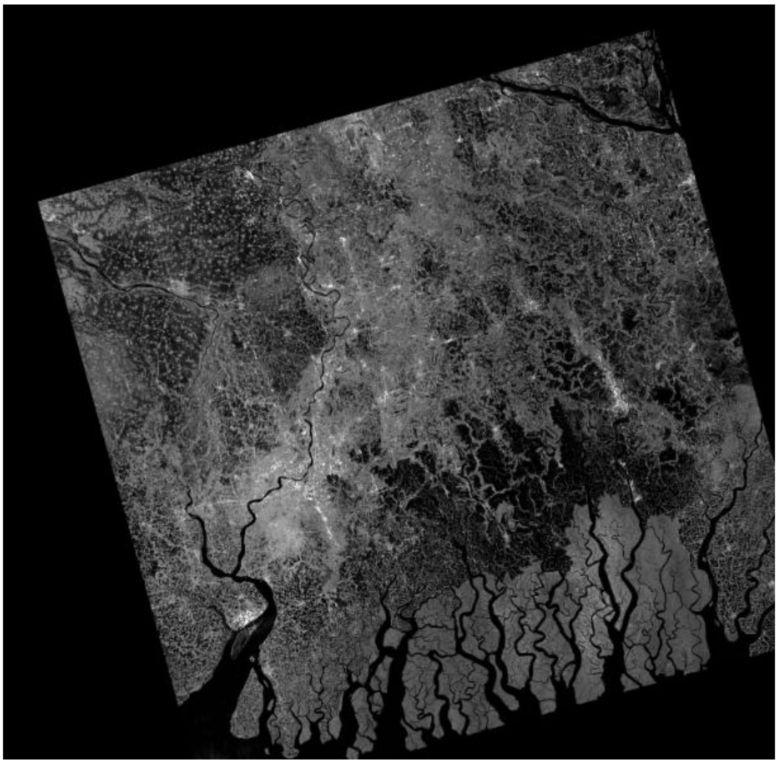

Figure 6.7: NISAR S-SAR Product Visualization

Source: Bhoonidhi Portal, NRSC/ISRO

6.8 Product Information and Metadata

After selecting a NISAR product, users can check its basic information. This information helps users understand the data before downloading it. Some important information includes:

- Product name
- Date and time
- Product type
- Polarization

- Area covered
- File size
- Processing information
- Metadata

This information is useful for selecting the correct data for analysis.

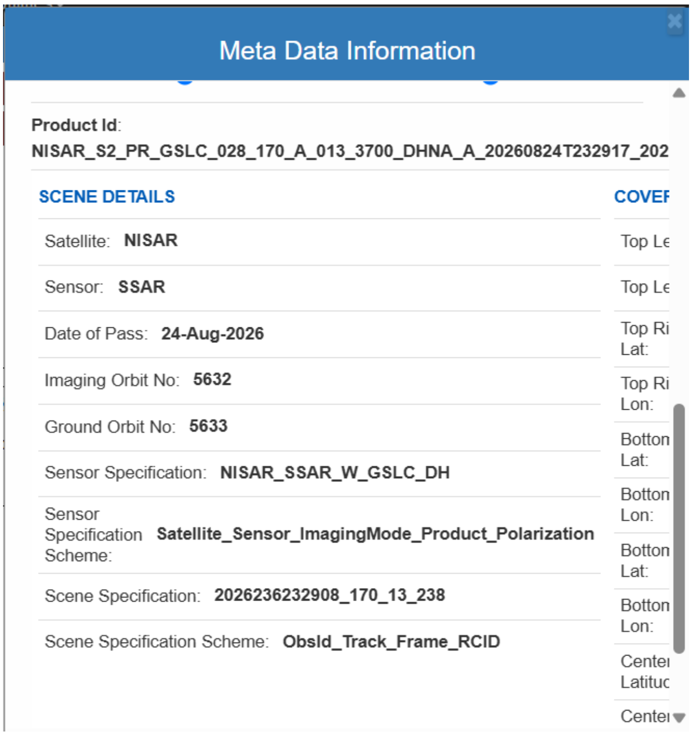

Figure 6.8: NISAR Product Information/Details

Source: Bhoonidhi Portal, NRSC/ISRO

6.9 Data Access and Download

After selecting the required product, users can access and download the data from the portal. The download option depends on the availability and type of data. The downloaded NISAR data can be used in software such as Python, QGIS and ArcGIS for further analysis.

6.10 Bhoonidhi Resources and Supporting Tools

The Resources section provides useful information and tools for working with satellite data. These resources can help users understand

and process NISAR data. The tools can also be useful for working with large HDF5 files and preparing the data for GIS analysis.

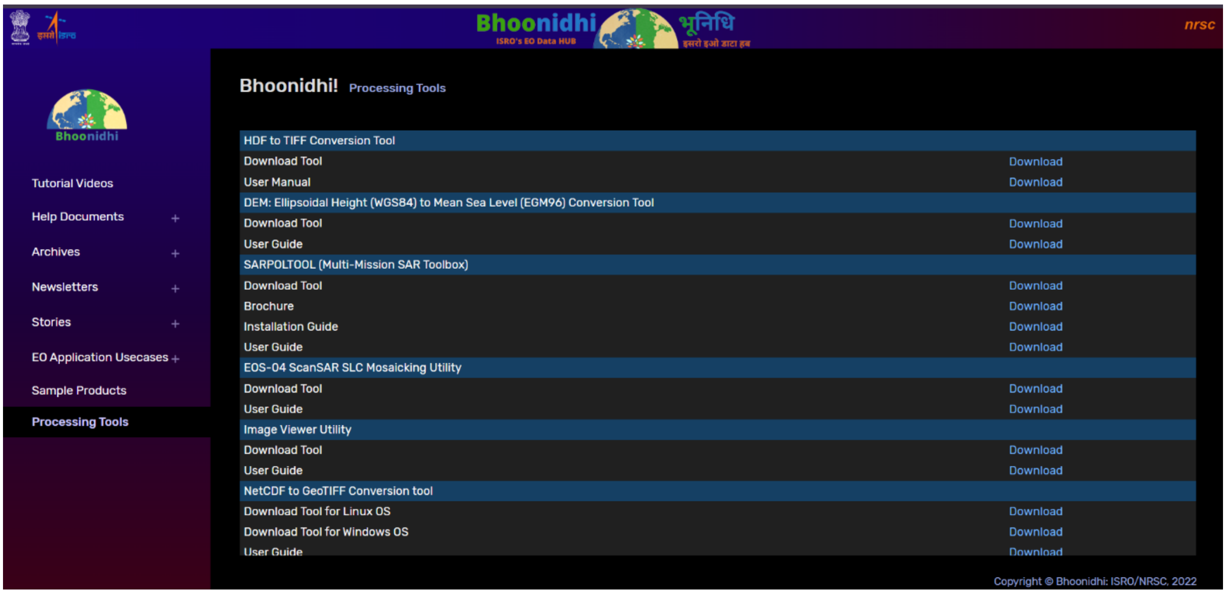

Figure 6.10: Bhoonidhi Processing Tools and Supporting Utilities

Source: Bhoonidhi Portal, ISRO/NRSC

6.11 Overall Observation from Bhoonidhi Portal

The Bhoonidhi Portal is useful for searching and accessing satellite data. Users can search data by place, satellite, sensor, resolution and product. The NISAR section provides information about NISAR S-band SAR data and its products. The portal also provides options to view product information, check the data and access the required dataset. Overall, Bhoonidhi is a useful platform for students, researchers and GIS and Remote Sensing users.

7. Downstream Applications of NISAR Data

NISAR data can be used for different Earth observation and geospatial applications. Some important downstream applications are:

7.1 Agriculture

NISAR data can help monitor crop fields, agricultural land and changes in vegetation.

7.2 Forest Monitoring
NISAR SAR data can be used to monitor forests, vegetation changes and forest cover.

7.3 Flood Mapping
SAR data can help identify flooded areas because it can collect information even under cloudy conditions.

7.4 Disaster Monitoring
NISAR data can support the monitoring of floods, landslides, earthquakes and other natural disasters.

7.5 Land Deformation
NISAR data can be used to study changes in the Earth's surface, such as land subsidence and tectonic movement.

7.6 Urban Monitoring
NISAR backscatter data can help identify built-up areas and monitor urban growth.

7.7 Water and Coastal Monitoring
NISAR data can be used to monitor rivers, wetlands, coastal areas and changes in water bodies.

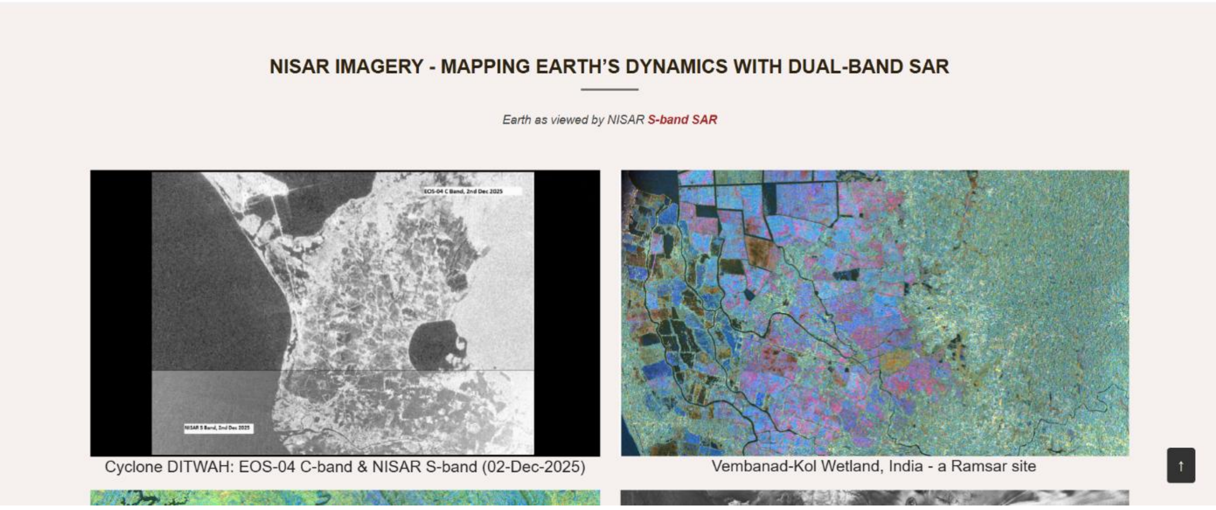

Figure 7.1: Examples of NISAR S-band SAR Applications in Earth Observation

Source: ISRO–NRSC, NISAR Mission / NISAR Imagery.

8. FUTURE RESEARCH OPPORTUNITIES

NISAR data can be useful for future research in different areas because of its repeated SAR observations and multi-frequency capabilities.

8.1 Artificial Intelligence and Machine Learning

NISAR data can be combined with AI and ML for land-use classification, crop mapping, flood detection and change detection.

8.2 Multi-temporal Change Detection

Repeated observations can help monitor urban growth, forest changes, agricultural activities and surface changes over time.

8.3 SAR Interferometry

NISAR data can support studies of land subsidence, landslides, earthquakes and infrastructure movement.

8.4 Multi-sensor Data Fusion

NISAR data can be combined with Sentinel-2 and Landsat data to improve land-cover and environmental studies.

8.5 Climate and Environmental Studies

NISAR can be used to study vegetation, wetlands, glaciers, soil and water-related changes.

8.6 Urban and Infrastructure Monitoring

NISAR observations can help monitor urban expansion, built-up areas and infrastructure development.**8.7 Disaster Management**

NISAR data can support flood mapping, landslide monitoring and earthquake-related surface change analysis.

9. PRACTICAL DEMONSTRATION USING NISAR DATA

9.1 Title of the Practical Study

NISAR S-band SAR Backscatter Analysis and Urban/Non-Urban Classification: A Case Study of Kolkata

9.2 Aim

The aim of this practical study was to use NISAR S-band SAR data for backscatter analysis and a simple urban/non-urban classification of the Kolkata study area. The practical work included backscatter visualization, conversion of backscatter values into decibel (dB), urban/non-urban classification, cleaning of the classification result, and preparation of a final overlay map.

9.3 Study Area

Kolkata was selected as the study area for this practical demonstration. A smaller Area of Interest (AOI) was selected from the available NISAR dataset to make the data processing and visualization easier. The selected area contains different land surface features, including built-up areas, vegetation, water bodies and other surface types. These different surfaces produce different radar backscatter responses.

9.4 NISAR Dataset

A NISAR S-band SAR dataset was used for this analysis. The selected data contained HHHH polarization information. The original NISAR dataset was available in HDF5 format. Since the complete dataset was very large, only a smaller spatial subset covering the Kolkata study area was used for the practical work. The required HHHH data was extracted and prepared for further analysis.

9.5 Methodology

The NISAR S-band HHHH data was extracted for the selected Kolkata study area and prepared in GeoTIFF format. The backscatter values were first visualized and then converted into decibel (dB). A simple threshold-based classification was used to separate the area into urban and non-urban classes. The classification result was cleaned to reduce isolated pixels, and finally, the cleaned urban classification was overlaid on the NISAR backscatter map.

Workflow:

NISAR Data → HHHH Extraction → GeoTIFF → dB Conversion → Urban/Non-Urban Classification → Cleaning → Final Overlay

9.6 Data Preparation

The NISAR dataset was first examined to identify the required HHHH polarization data. A spatial subset was created for the selected Kolkata study area. The extracted data was then converted into GeoTIFF format so that it could be easily used in GIS software for visualization and analysis.

Workflow:

NISAR HDF5 Data → HHHH Backscatter Extraction → Spatial Subset → GeoTIFF Conversion → dB Conversion → Urban/Non-Urban Classification → Cleaning → Final Overlay

9.7 NISAR SAR Backscatter Analysis

The extracted HHHH dataset was used to represent the radar backscatter of the study area. Radar backscatter shows how strongly different surface features reflect the incoming radar signal. Different land surfaces can produce different backscatter values because of differences in surface roughness, structure, moisture and land cover. The original backscatter map shows the spatial variation of NISAR HHHH backscatter across the Kolkata study area.

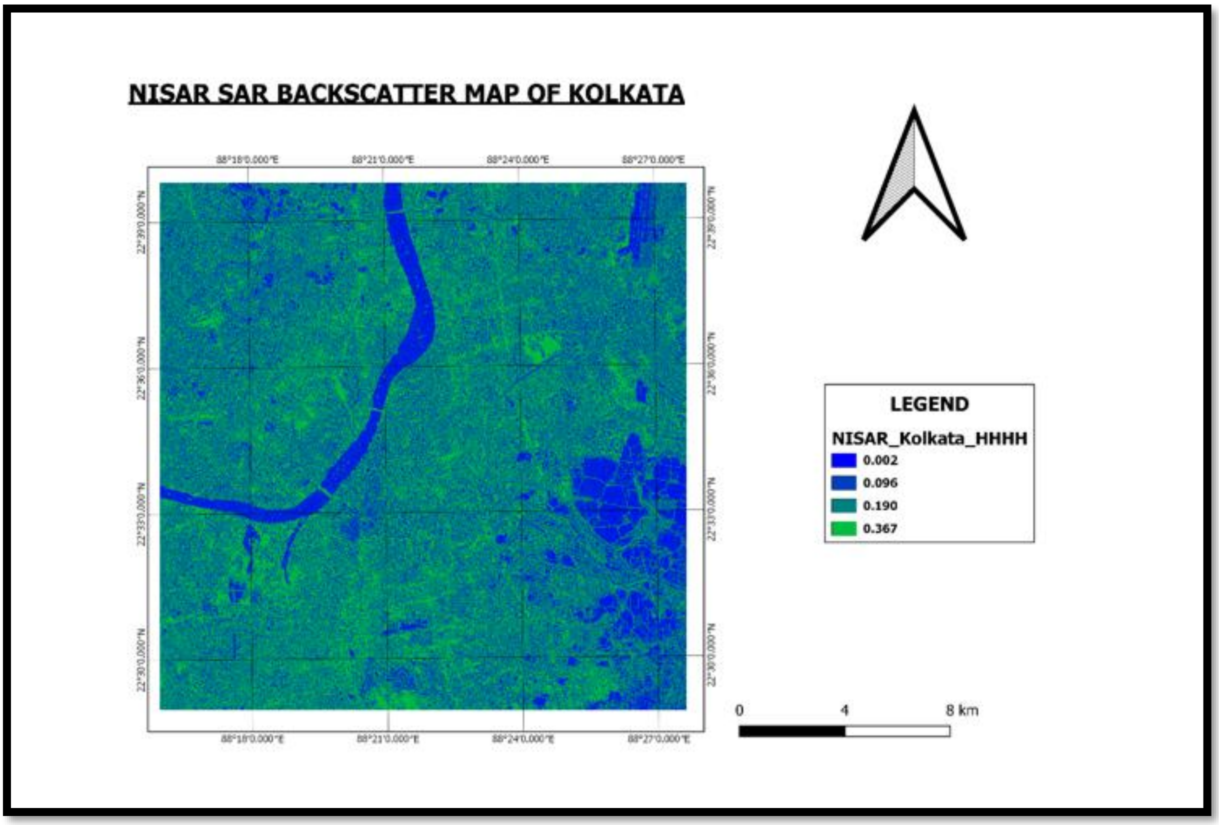

Figure 9.1: NISAR HHHH SAR Backscatter Map of the Kolkata Study Area

9.8 Conversion of Backscatter to Decibel (dB)

The original linear backscatter values were converted into decibel (dB) units for easier visualization and interpretation.

The following standard equation was used:

Backscatter (dB) = 10 × log10(Backscatter linear)

After the conversion, the HHHH backscatter raster was visualized as a dB map of the Kolkata study area. The dB conversion made the variation in radar backscatter easier to observe and compare across different parts of the study area. Areas with different surface characteristics showed different backscatter responses. The final dB map therefore provides a clear representation of the spatial variation of NISAR SAR backscatter across Kolkata.

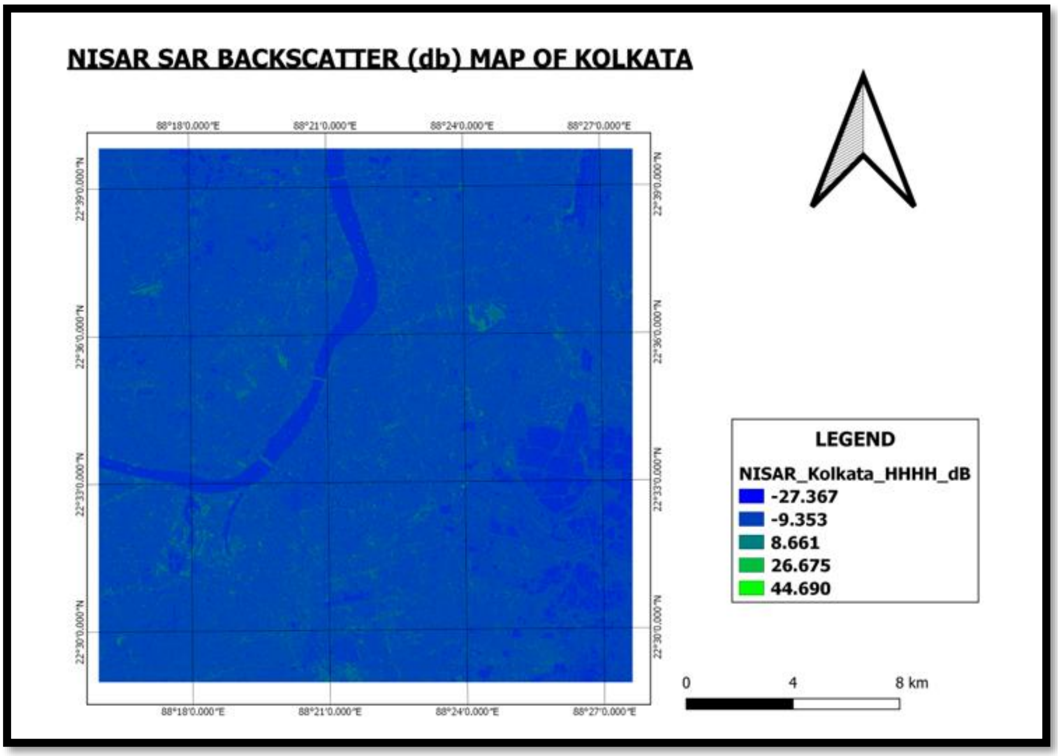

Figure 9.2: NISAR HHHH SAR Backscatter Map in Decibel (dB)

9.9 Urban/Non-Urban Classification and Cleaning

A simple binary classification was performed using the processed NISAR backscatter data.

Two classes were created:

- **Class 0 – Non-Urban Area**
- **Class 1 – Urban Area**

The classification was based on the difference in radar backscatter response between different surface types. Areas identified as urban were assigned the value **1**, while the remaining areas were assigned the value **0**. The first classification result contained some small and isolated pixels. Therefore, a cleaning process was applied to improve the visual appearance of the classified map and reduce unwanted isolated pixels.

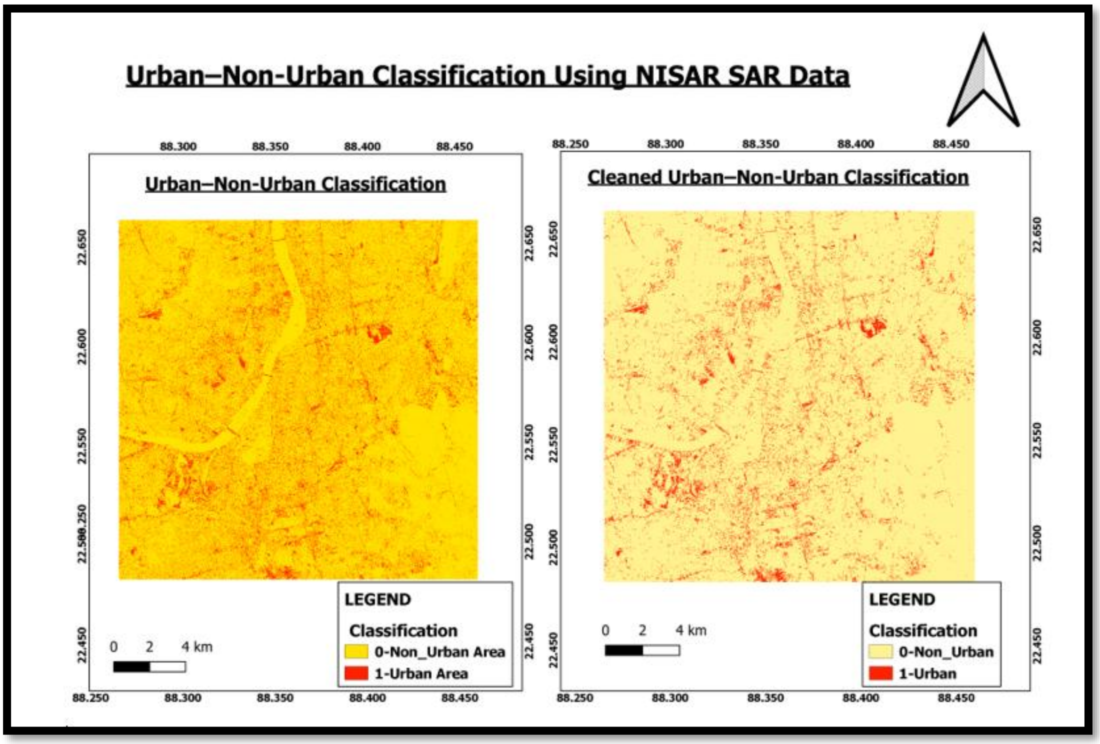

Figure 9.3: Initial and Cleaned Urban/Non-Urban Classification Using NISAR SAR Data

9.10 Final NISAR Backscatter and Urban Classification Overlay

The cleaned urban/non-urban classification was combined with the NISAR HHHH backscatter map to create the final overlay map. This final map was prepared to bring together the radar backscatter information and the urban classification result in a single visual representation. In the final map, the NISAR backscatter data in dB was used as the background layer, while the cleaned urban areas were displayed over it with transparency. This makes it easier to visually compare the classified urban areas with the underlying radar backscatter pattern. It also helps to understand how the areas identified as urban are related to different levels of SAR backscatter.

The final overlay provides a clear combined view of both the original radar information and the classification result. It helps in observing the spatial distribution of urban areas while also showing the variation of NISAR HHHH backscatter across the Kolkata study area.

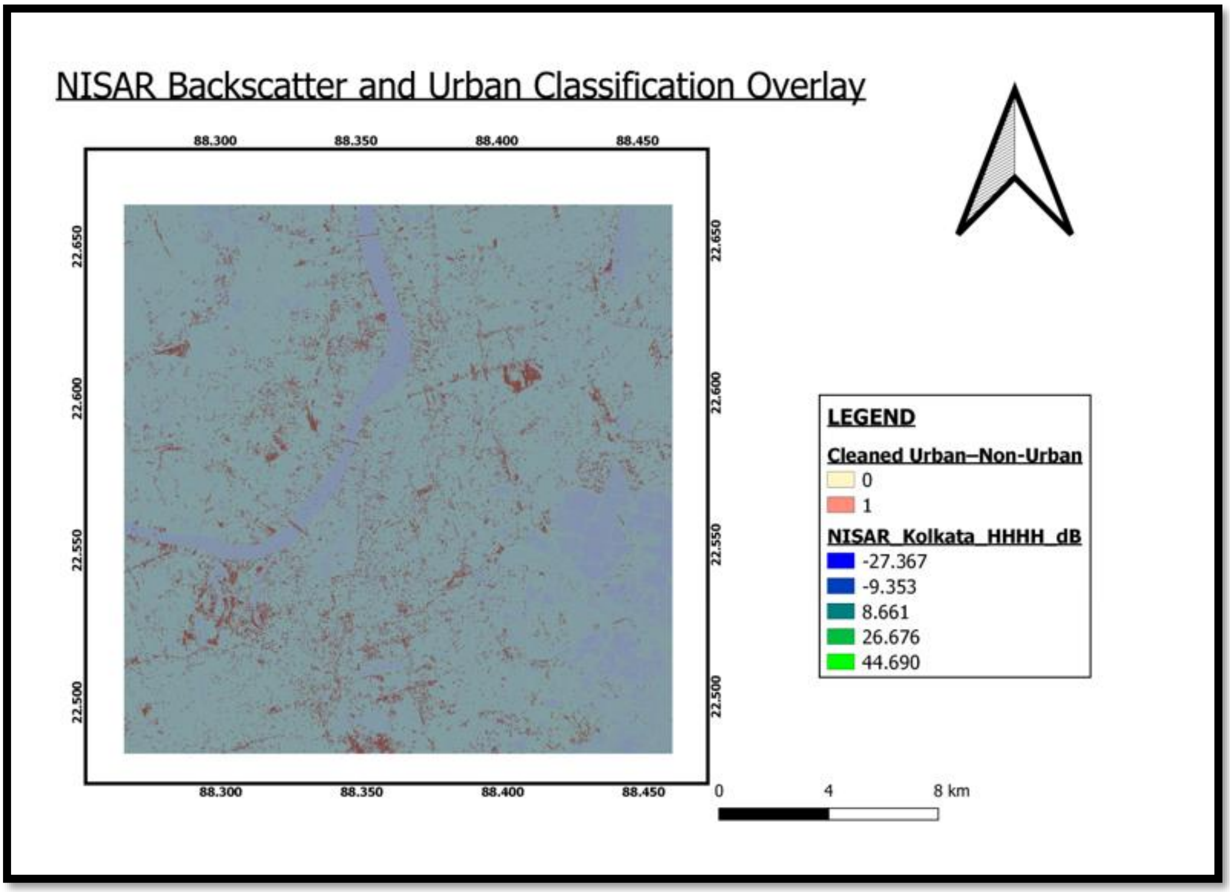

Figure 9.4: Final Overlay of NISAR HHHH Backscatter and Urban Classification

9.11 Results and Observations

The practical study showed a simple way to process NISAR S-band SAR data for urban mapping. First, the HHHH backscatter data was extracted and displayed as a map. Then, the original backscatter values were converted into dB for easier viewing and understanding. After that, the processed data was used to classify the study area into urban and non-urban areas. The first classification map had some small and isolated pixels. Therefore, the classification was cleaned to make the result clearer. Finally, the cleaned urban classification was combined with the NISAR backscatter map to create the final overlay map.

The four maps show the complete process:

1. **Original NISAR SAR Backscatter Map**
2. **NISAR SAR Backscatter Map in dB**

3. **Initial and Cleaned Urban/Non-Urban Classification**
4. **Final NISAR Backscatter and Urban Classification Overlay**

Overall, the results show that NISAR S-band SAR data can be processed and used for a simple urban mapping study. Using a smaller Area of Interest also made the large NISAR dataset easier to process, display and analyse.

9.12 Limitations

The urban/non-urban classification used in this study is a simple method, so some errors may be present in the result.

SAR backscatter can change because of different factors such as surface roughness, moisture, vegetation and building structure. Because of this, some non-urban areas may sometimes show backscatter values similar to urban areas.

The classification was based on only two classes, urban and non-urban. It was also not checked using ground-truth or reference data. Therefore, this result should be considered as a simple practical demonstration and not a fully accurate urban classification.

9.13 Future Improvement

The practical work can be improved in the future by:

- Using multiple NISAR observations
- Using additional SAR polarizations
- Using multi-temporal NISAR data
- Combining NISAR SAR data with Sentinel-2 optical data
- Applying machine learning methods such as Random Forest or SVM
- Using ground-truth or reference data
- Performing classification accuracy assessment

- Conducting urban change detection using multi-temporal data

10. GITHUB REPOSITORY AND DOCUMENTATION

The practical work can be documented in a GitHub repository. The repository can contain the code, maps, results and other documents related to the project. The processing steps and screenshots can also be included so that the work is easy to understand. The final maps and results can be stored in separate folders. The original NISAR dataset is very large, so it may not be uploaded directly to GitHub. Instead, information about the data source and data processing steps can be provided.

The repository can have the following folders:

- **code/** – for the processing code
- **documentation/** – for methodology and data details
- **results/** – for the final maps
- **screenshots/** – for screenshots
- **data/** – for information about the NISAR dataset

A **README.md** file can also be included. It can give a short description of the project, including the aim, study area, dataset, methodology and results.

11. CONCLUSION

This study provided a basic understanding of the NISAR mission and its data products. It also helped to understand how NISAR data can be searched, accessed and used for practical applications. In the practical work, NISAR S-band HHHH backscatter data was used for the Kolkata study area. The backscatter data was processed, converted into dB and displayed as a map. A simple urban/non-urban classification was then created. The final overlay map showed the urban classification together with the NISAR backscatter data.

Overall, the study shows that NISAR SAR data can be useful for remote sensing and GIS applications. In the future, the work can be improved by using more data, different polarizations and advanced classification methods.

12.References

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4. **NISAR Science and Applications** — NASA/ISRO.
5. **QGIS Documentation** — QGIS Geographic Information System.
6. **ArcGIS Pro Documentation** — Esri.